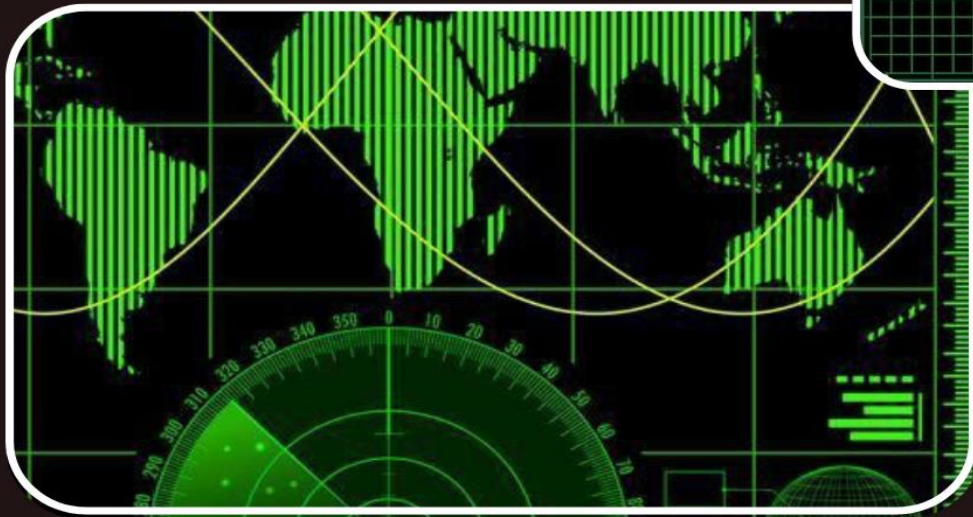
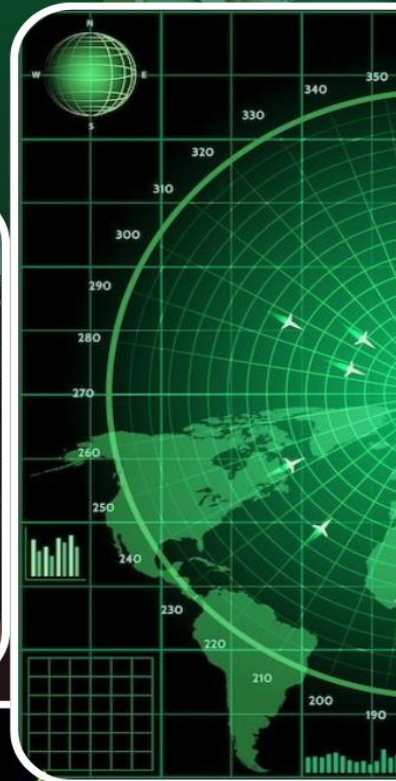


RADAR SYSTEM

- object Detection
- Range Measurement
- Speed Detection
- High Resolution
- Clutter Rejection



RADAR SYSTEM USING ARDUINO AND ULTRASONIC SENSOR

Abstract

The Radar System using Arduino and an Ultrasonic Sensor is a simple yet effective project demonstrating how technology can detect objects, measure distances, and visualize the surroundings. In this system, we use an ultrasonic sensor to send sound waves, which bounce back when they hit an object. By measuring the time, it takes for the waves to return, we can calculate the object's distance. The system is controlled by an Arduino microcontroller, which processes the sensor data and controls the sensor's rotation using a servo motor.

This radar system is designed to detect obstacles in a particular area by rotating the ultrasonic sensor across different angles. The servo motor allows the sensor to rotate back and forth, scanning a wide area. The Arduino reads the time it takes for the ultrasonic waves to return and calculates the distance to the nearest object. This distance is then displayed through a connected computer or displayed on a simple graphical user interface (GUI). The radar system is useful for applications such as robotics, where it can be used for obstacle avoidance, or in security systems, where it can detect intruders.

One of the main advantages of this system is that it is low-cost, simple to build, and energy efficient. The system also provides real-time data, making it suitable for various practical applications. By using an open-source platform like Arduino, the project is easily customizable and can be enhanced further with additional sensors or features. The overall goal of this project is to provide a clear, hands-on understanding of how radar technology works using affordable and accessible hardware.

Introduction

Radar (Radio Detection and Ranging) is a technology that uses electromagnetic waves to detect objects, and measure their distance, speed, and direction. It is widely used in various fields, such as navigation, weather forecasting, military defense, and even in autonomous vehicles. The principle behind radar systems is based on sending out waves (either electromagnetic or sound waves) that bounce off objects, and by analyzing the reflection, the system can determine the position and distance of the object.

In this project, we aim to develop a simple and low-cost radar system using an Arduino microcontroller and an ultrasonic sensor. The ultrasonic sensor works by emitting sound waves that travel through the air. When these waves hit an object, they bounce back to the sensor, and by measuring the time taken for the waves to return, the system can calculate the distance to the object. To increase the area covered by the sensor, a servo motor is used to rotate the sensor in a specific range, scanning different angles to detect objects in various directions.

This radar system can be used in many practical applications, such as in robotics for obstacle detection and avoidance, in security systems to detect intruders, or in other areas where detecting objects and measuring their distance is essential. The system's simplicity and the use of affordable components make it an accessible way to understand and apply basic radar principles. By utilizing Arduino, which is an open-source platform, this project becomes customizable, allowing users to tweak the system to suit their specific needs.

The main focus of this system is to demonstrate how radar technology works in a hands-on way, using easily available and cost-effective components. It highlights the power of combining hardware and software to create functional devices that can solve real-world problems.

Design and Implementation of Radar System

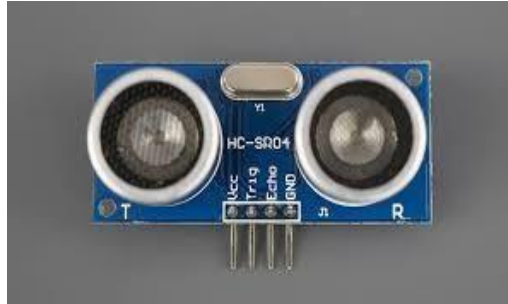
The radar system designed using Arduino and ultrasonic sensors can be implemented without using a breadboard by directly connecting the components to the Arduino and arranging them on a fixed platform. This section outlines the design and implementation steps, focusing on the components, their interconnections, and the overall system setup without the need for a breadboard.

A) Hardware System Design

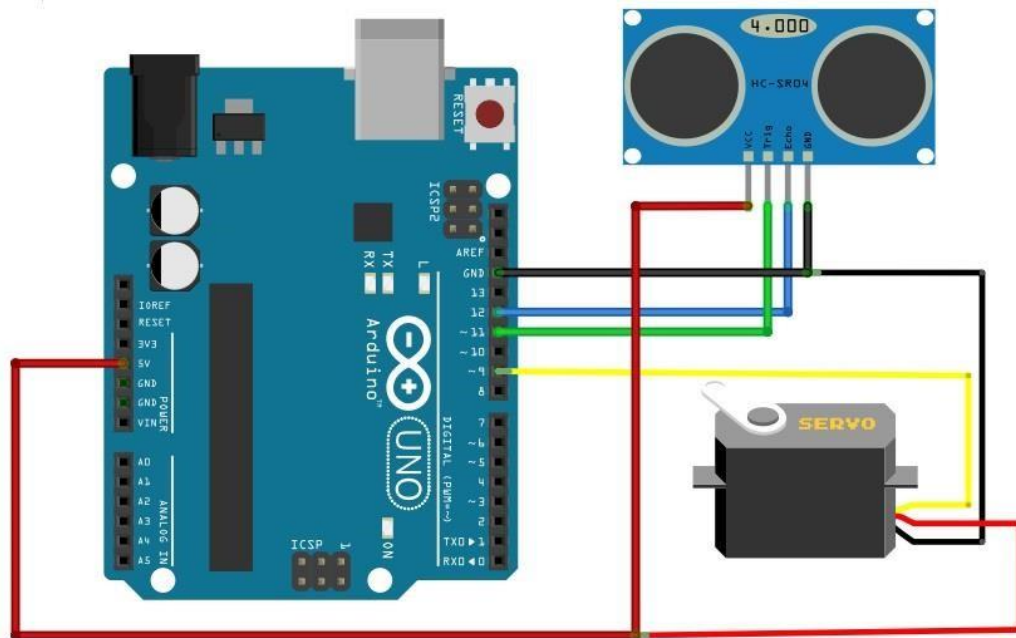
The primary components involved in the hardware system are:

1. **Arduino Microcontroller:** The central controller of the system, responsible for processing input from the ultrasonic sensor and controlling the servo motor. It also handles the data collection and output.
2. **Ultrasonic Sensor (HC-SR04):** This sensor is responsible for sending out ultrasonic pulses and receiving the reflected echoes from objects in the environment. By measuring the time taken for the sound waves to return, the distance to the detected objects is calculated.
3. **Servo Motor (SG90):** The servo motor is used to rotate the ultrasonic sensor across a range of angles. This motion allows the sensor to scan different areas and detect obstacles in various directions.





4. **Wires for Direct Connections:** Instead of using a breadboard, all components are directly connected using jumper wires, with careful attention to the pin out and wiring.



B) Component Connections

Without using a breadboard, each component is connected directly to the Arduino microcontroller using jumper wires. The connections are made as follows:

- **Ultrasonic Sensor:**
 - The **Trig Pin** of the ultrasonic sensor is connected to a digital pin on the Arduino (e.g., pin 8).
 - The **Echo Pin** of the ultrasonic sensor is connected to another digital pin on the Arduino (e.g., pin 7).
 - The **VCC Pin** of the ultrasonic sensor is connected to the 5V pin of the Arduino.
 - The **GND Pin** of the ultrasonic sensor is connected to the GND pin of the Arduino.
- **Servo Motor:**
 - The **Control Pin** (signal wire) of the servo motor is connected to a digital pin on the Arduino (e.g., pin 6).

- The **VCC Pin** of the servo motor is connected to the 5V pin of the Arduino. ○
The **GND Pin** of the servo motor is connected to the GND pin of the Arduino.
- **Power Supply:** The Arduino itself is powered through its USB connection or an external power supply. The ultrasonic sensor and servo motor are powered directly from the 5V pin on the Arduino, ensuring that all components are correctly supplied with power.

C) Mechanical Assembly and Setup

Instead of using a breadboard to mount the components, they are arranged on a fixed base or platform. The ultrasonic sensor is securely mounted onto the servo motor. The servo motor is then placed on a stable surface where it can rotate freely.

This setup ensures that the servo motor can rotate the sensor to different angles (usually between 15° and 165°), allowing the radar system to scan its environment. The Arduino board is typically placed near the other components to ensure short, direct connections using jumper wires.

D) Control Logic and Software

The control logic of the system remains the same as when using a breadboard. The Arduino controls the servo motor's rotation and the ultrasonic sensor's pulse emission. The program cycles through different angles for the servo motor, triggering the ultrasonic sensor to emit sound waves and measure the time it takes for the waves to return.

The Arduino program handles the following tasks: • It controls the movement of the servo motor across the defined range of angles.

- At each position, it sends a pulse to the ultrasonic sensor, waits for the echo, and calculates the distance.
- The distance value is then sent to the serial monitor or GUI for visualization.

E) System Integration and Testing

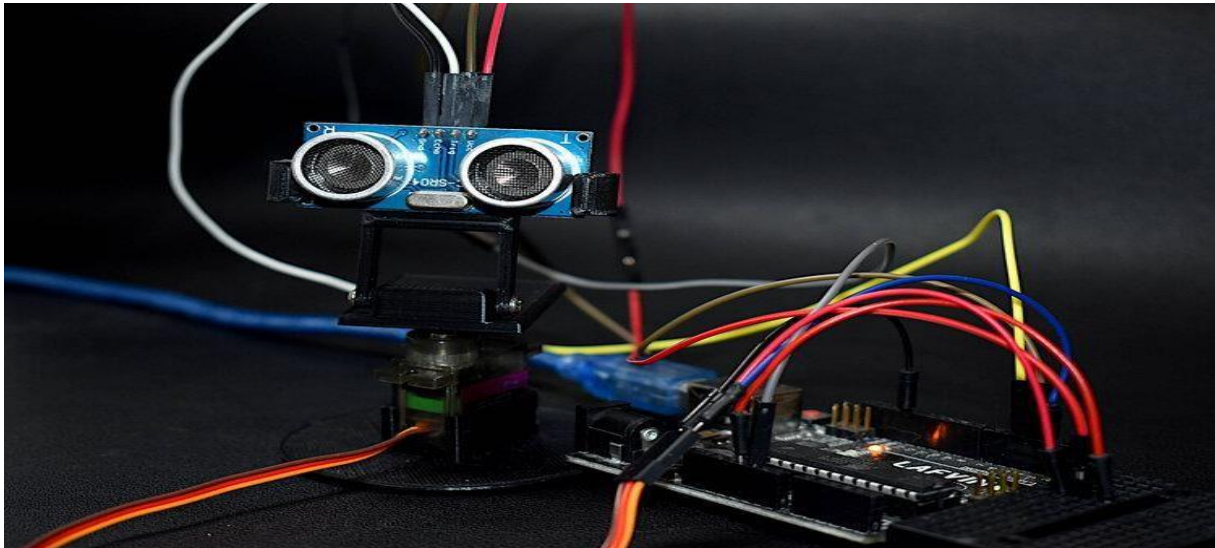
In this setup, after the components are connected directly to the Arduino, the system is powered up. The ultrasonic sensor starts emitting pulses as the servo motor rotates, scanning for obstacles in the radar's field of view.

System testing involves verifying that the ultrasonic sensor accurately detects objects and calculates distances at various angles. The output is displayed on the Arduino serial monitor or a connected GUI for visualization.

F) GUI System Design and Output

The graphical user interface (GUI) receives the distance and angle data from the Arduino. The GUI is typically designed to show a visual representation of the radar system's operation, such as a rotating line representing the radar's sweep, with smudges or marks indicating detected objects.

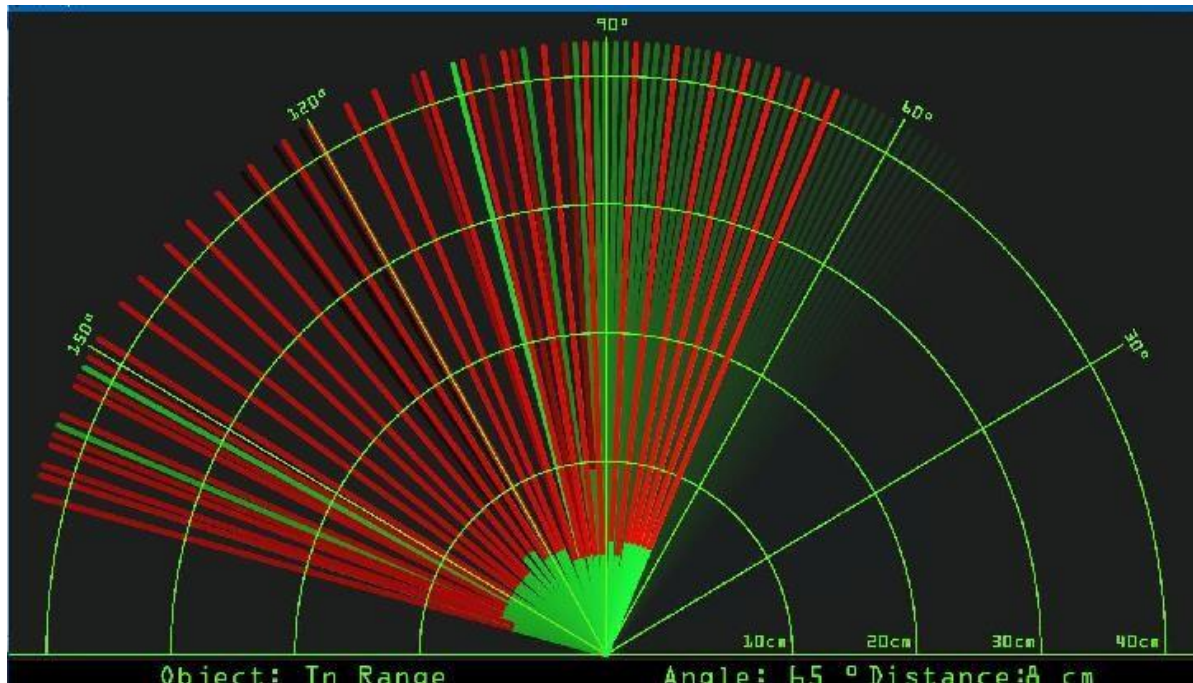
The GUI may be developed using Java, Python, or other programming languages, and it can display the radar scan in real time as the sensor moves across the environment. The interface may also include additional features, such as data logging or visualization of the radar's field of view.



G) Final Testing and Optimization

Once the radar system is set up and integrated, final testing is performed to ensure its accuracy and reliability. The system should be tested in different environments to evaluate its performance in detecting obstacles and calculating distances. If needed, the software or hardware configuration is adjusted for optimal performance.

For example, the rotation speed of the servo motor may be adjusted for smoother operation, or the ultrasonic sensor's range may be calibrated based on the testing conditions.



ARDUINO CODE:

```
// Includes the Servo library

#include <Servo.h>

// Defines Trig and Echo pins of the Ultrasonic Sensor

const int trigPin = 10; const int echoPin = 11;

// Variables for the duration and the distance

long duration; int distance;

Servo myServo; // Creates a servo object for controlling the servo motor

void setup() {  pinMode(trigPin, OUTPUT); // Sets the trigPin as an
Output  pinMode(echoPin, INPUT); // Sets the echoPin as an Input

Serial.begin(9600);  myServo.attach(9); // Defines on which pin is the
servo motor attached

    // rotates the servo motor from 15 to 165 degrees

// Calls a function for calculating the distance measured by the Ultrasonic sensor for each degree
```

```
// Sends the current degree into the Serial Port

// Sends addition character right next to the previous value needed later in the Processing IDE for
indexing

// Sends the distance value into the Serial Port

// Sends addition character right next to the previous value needed later in the Processing IDE for
indexing

// Repeats the previous lines from 165 to 15 degrees

// Function for calculating the distance measured by the Ultrasonic sensor int

// Sets the trigPin on HIGH state for 10 micro seconds

// Reads the echoPin, returns the sound wave travel time in microseconds
```

PROCESSING CODE:

```
import processing.serial.*; // imports library for serial communication import
java.awt.event.KeyEvent; // imports library for reading the data from the serial port import
java.io.IOException;

Serial myPort; // defines Object Serial

// defubes variables

String angle="";

String distance="";

String data="";

String noObject;

float pixsDistance;

int iAngle, iDistance;

int index1=0; int
index2=0; PFont

// ***CHANGE THIS TO YOUR SCREEN RESOLUTION***
```



```
// starts the serial communication

// reads the data from the serial port up to the character '!'. So actually, it reads this: angle,distance.

// simulating motion blur and slow fade of the moving line

// green color

// calls the functions for drawing the radar

// starts reading data from the Serial Port

// reads the data from the Serial Port up to the character '!' and puts it into the String variable "data".


// converts the String variables into Integer

// draws the angle lines

// moves the starting coordinats to new location

// red color

// covers the distance from the sensor from cm to pixels

// limiting the range to 40 cms

// draws the object according to the angle and the distance

moves the starting coordinats to new location line

// draws the line according to the angle
```

Conclusion

This radar system, designed using Arduino and ultrasonic sensors, can operate without a breadboard, offering a compact and efficient solution. The system effectively detects objects in its path by emitting ultrasonic waves and measuring the reflections. The data collected is then processed and displayed in real-time on a GUI or serial monitor. This implementation demonstrates a practical and cost-effective approach to building a basic radar system for object detection and obstacle avoidance.